

Lesson 4: subspaces and the basis for a subspace

Linear subspaces

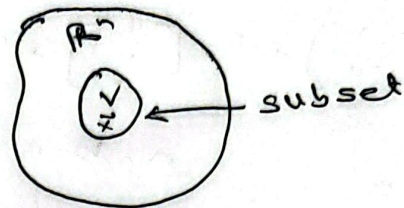
Subspace of $\mathbb{R}^n \leftarrow \left\{ \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix} \mid x_i \in \mathbb{R} \ 1 \leq i \leq n \right\}$
 $V \leftarrow$ subset of $\mathbb{R}^n$

V is subspace of $\mathbb{R}^n$

$\Rightarrow V$ contains $0 = \begin{bmatrix} 0 \\ \vdots \\ 0 \end{bmatrix}$

$\Rightarrow \vec{x} \in V \Rightarrow c\vec{x} \in V$

closure under scalar multiplication



$\Rightarrow$ Subspace.

$\Rightarrow \vec{a} \in V \quad \vec{a} + \vec{b} \in V$
 $\vec{b} \in V$

closure under addition

$$V = \{0\} = \left\{ \begin{bmatrix} 0 \\ 0 \\ \vdots \\ 0 \end{bmatrix} \right\}$$

V a subspace of $\mathbb{R}^3$
 zero vector $\checkmark$

$$c \begin{bmatrix} 0 \\ 0 \\ \vdots \\ 0 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ \vdots \\ 0 \end{bmatrix} \quad \text{closed under multiplication } \checkmark$$

$$\begin{bmatrix} 0 \\ 0 \\ \vdots \\ 0 \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \\ \vdots \\ 0 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ \vdots \\ 0 \end{bmatrix} \quad \text{closed under addition } \checkmark$$

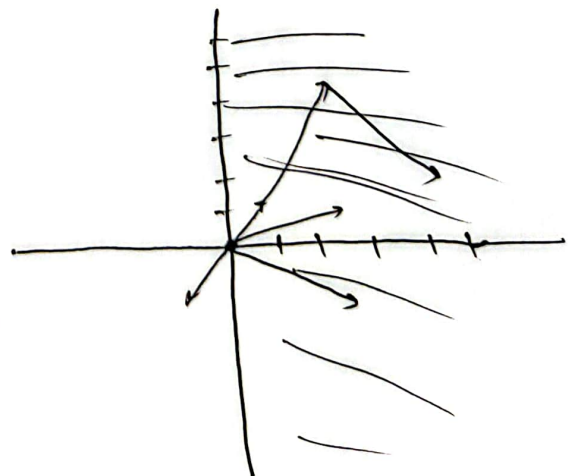
$$S = \left\{ \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} \in \mathbb{R}^2 \mid x_i \geq 0 \right\}$$

is S subspace of $\mathbb{R}^2$

contain $\begin{bmatrix} 0 \\ 0 \end{bmatrix} \checkmark$

$$\begin{bmatrix} a \\ b \end{bmatrix} + \begin{bmatrix} c \\ d \end{bmatrix} = \begin{bmatrix} a+c \\ b+d \end{bmatrix} \rightarrow \begin{matrix} \geq 0 \checkmark \\ \text{closed under addition } \checkmark \end{matrix}$$

$$\neg \begin{bmatrix} a \\ b \end{bmatrix} = \begin{bmatrix} -a \\ -b \end{bmatrix} \rightarrow \begin{matrix} \text{negative} \\ \text{X} \end{matrix}$$



subset of S is not subspace of $\mathbb{R}^2$ as not closed under scalar multiplication.

$$U = \text{span}(v_1, v_2, v_3) \quad \text{is a valid subspace of } \mathbb{R}^n$$

$$0\vec{v}_1 + 0\vec{v}_2 + 0\vec{v}_3 = \vec{0} \quad \checkmark$$

$$\vec{x} = c_1\vec{v}_1 + c_2\vec{v}_2 + c_3\vec{v}_3$$

$$a\vec{x} = a(c_1\vec{v}_1 + c_2\vec{v}_2 + c_3\vec{v}_3) = \underbrace{ac_1}_{c_4}\vec{v}_1 + \underbrace{ac_2}_{c_5}\vec{v}_2 + \underbrace{ac_3}_{c_6}\vec{v}_3 \quad \leftarrow \text{closed under multiplication } \checkmark$$

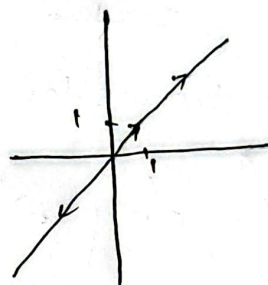
$$\vec{y} = d_1\vec{v}_1 + d_2\vec{v}_2 + d_3\vec{v}_3$$

$$\vec{x} + \vec{y} = (c_1 + d_1)\vec{v}_1 + (c_2 + d_2)\vec{v}_2 + (c_3 + d_3)\vec{v}_3 \quad \leftarrow \text{closed under addition } \checkmark$$

$$U = \text{span}\left(\begin{bmatrix} 1 \\ 1 \end{bmatrix}\right) \quad \text{or } \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

$$0\begin{bmatrix} 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix} \quad \checkmark$$

$$c_1\begin{bmatrix} 1 \\ 1 \end{bmatrix} + c_2\begin{bmatrix} 1 \\ 1 \end{bmatrix} = \underbrace{(c_1 + c_2)}_{\text{arbitrary constant}} \begin{bmatrix} 1 \\ 1 \end{bmatrix} \quad \checkmark$$



Basis of a subspace

$$V = \text{span}(\vec{v}_1, \vec{v}_2, \dots, \vec{v}_n)$$

$$c_1\vec{v}_1 + c_2\vec{v}_2 + \dots + c_n\vec{v}_n \quad c_i \in \mathbb{R}$$

↑ subspace $\{v_1, v_2, \dots, v_n\}$ linearly independent

$$c_1\vec{v}_1 + c_2\vec{v}_2 + \dots + c_n\vec{v}_n = \vec{0} \quad \vec{v}_1 + \vec{v}_2$$

$$c_1 = c_2 = \dots = c_n = 0$$

$$S = \{\vec{v}_1, \vec{v}_2, \dots, \vec{v}_n\}$$

$$T = \{\vec{v}_1, \vec{v}_2, \dots, \vec{v}_n, \underbrace{\vec{v}_c}_{\text{circled}}\}$$

S is a basis for V

$$\text{span}(T) = V$$

Basis "minimum set of vectors that spans the subspace"

T is a linearly dependent
 T is not a basis for V

$$S = \left\{ \begin{bmatrix} 2 \\ 3 \end{bmatrix}, \begin{bmatrix} 7 \\ 0 \end{bmatrix} \right\} \quad \text{span}(S) \subseteq \mathbb{R}^2 \quad \text{span}(S) = \mathbb{R}^2$$

$$x_1, x_2 \in \mathbb{R}$$

$$c_1 \begin{bmatrix} 2 \\ 3 \end{bmatrix} + c_2 \begin{bmatrix} 7 \\ 0 \end{bmatrix} = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$$

S is also linearly independent

$$2c_1 + 7c_2 = x_1$$

$$2/3 x_2 + 7c_2 = x_1$$

$$3c_1 + 0 = x_2$$

$$\Rightarrow c_1 = \frac{x_2}{3}$$

$$7c_2 = x_1 - 2/3 x_2$$

$$c_2 = \frac{x_1}{7} - \frac{2}{21} x_2$$

$$c_1 \begin{bmatrix} 2 \\ 3 \end{bmatrix} + c_2 \begin{bmatrix} 7 \\ 0 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$$x_1 = 0$$

$$x_2 = 0$$

$$c_1 = 0$$

$$c_2 = 0$$

∴ also

S is a basis for $\mathbb{R}^2$.

$$T = \left\{ \begin{bmatrix} 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \end{bmatrix} \right\} \xrightarrow{\text{standard basis}} \text{span } \mathbb{R}^2 \rightarrow \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$$

T is also a basis for $\mathbb{R}^2$.

$$x_1 \begin{bmatrix} 1 \\ 0 \end{bmatrix} + x_2 \begin{bmatrix} 0 \\ 1 \end{bmatrix} = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} \quad \text{linearly independent}$$

$\{v_1, v_2, \dots, v_n\} = \text{Basis for } U \leftarrow \text{subspace}$

$$\vec{a} \in U \quad \vec{a} = c_1 \vec{v}_1 + c_2 \vec{v}_2 + \dots + c_n \vec{v}_n$$

$$\vec{a} = d_1 \vec{v}_1 + d_2 \vec{v}_2 + \dots + d_n \vec{v}_n$$

$$\vec{0} = (c_1 - d_1) \vec{v}_1 + (c_2 - d_2) \vec{v}_2 + \dots + (c_n - d_n) \vec{v}_n$$